



CSEE 4290: Computer Architecture

Top Level SCC – Spring 2025 – Group 5

[https://github.com/Herring-UGACSEE-4290-F25/Group-5/tree/main/Proj2\(SCC\)/SCCPrefetchBPExeCheckCompleted](https://github.com/Herring-UGACSEE-4290-F25/Group-5/tree/main/Proj2(SCC)/SCCPrefetchBPExeCheckCompleted)

Instruction Fetch

Decode

Execute

Register File

Fix Unconditional Branch Bubble

To-Do

Write Unique Test Case

Create Testbench

Implement uCode MUL/S

Group Responsibilities

Test Bench: Nathaniel

Instruction Fetch: Ricky

Instruction Decode: Matthew

Execute: Nathaniel

uCode MUL/S: Everybody

Schedule

ID/IF: 10-03

Execute 3 Instructions/Branch Imm: 10-10

Execute Complete: 10-20

Testcases: 10-24

SCC Complete: 10-29

Testcase Plan

- Checks LOAD and STORE
- Checks ADD
- Checks SUBS
- Checks B and B.cond
- Checks HALT